

2(a)	(angular frequency =) $2\pi \times$ frequency or $2\pi/\text{period}$	B1
2(b)(i)	1. displacement = 2.0 cm	A1
	2. amplitude = 1.5 cm	A1
2(b)(ii)	reference to displacement of oscillations or displacement from equilibrium position or displacement from 2.0 cm	B1
	straight line indicates acceleration $\propto$ displacement	B1
	negative gradient shows acceleration and displacement are in opposite directions	B1

Question	Answer	Marks
2(b)(iii)	$\omega^2 = (-)1/\text{gradient}$ or $\omega^2 = (-)\Delta a/\Delta s$ or $a = (-)\omega^2 x$ <u>and</u> correct value of x	C1
	= e.g. (1.8 / 0.03) or (0.9 / 0.015) or (1.2 / 0.02) etc. or $0.9 = \omega^2 \times 0.015$ = 60	C1
	$f = \sqrt{60/2\pi}$ = 1.2 Hz	A1

Question	Answer	Marks
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